PROJECT 1: LINEAR REGRESSION

MASM22/FMSN30/FMSN30F/FMSN40: LINEAR AND LOGISTIC REGRESSION (WITH DATA GATHERING), 2026

Peer assessment version: **17.00 on Monday 27 April**

Peer assessment comments: **17.00 on Tuesday 28 April**

Final version: **17.00 on Wednesday 29 April**

Introduction

We want to model the yearly emissions of atmospheric particles with a diameter between 2.5 and 10 µm, PM_{10}, per capita in Sweden's 290 municipalities using data from Statistics Sweden (Statistiska Centralbyrån), `www.scb.se`. All data is from 2021 and located in `Project1data.xlsx`. See Canvas for maps of the locations of municipalities and counties.

Variable namn	Description
`Kommun`	Municipality number and name
`County`	County (Län) number and name
`Part`	Part of Sweden: 1 = Götaland; 2 = Svealand; 3 = Norrland
`Coastal`	Coastal = any sea area within its borders: 0 = Inland; 1 = Coastal
`Fuel`	Yearly fuel consumption in agriculture, forestry, fishing, production, building, public, transports, and other sectors (GWh / 1000 inhabitants)
`Vehicles`	Number of passenger cars, busses and trucks / 1000 inhabitants
`Builton`	Area covered in buldings, roads, etc (= not nature), (hectares / 1000 inhabitants)
`Seniors`	65+ year olds (percentage)
`Higheds`	At least 3 years of post-secondary (eftergymnasial) education (percentage)
`Income`	Median yearly income (1000 SEK)
`GRP`	Gross Regional Product per capita (1000 SEK)
`PM10`	The yearly emission of PM_{10}-particles (metric tonnes / 1000 inhabitants)

Our goal is to model how the yearly emission of PM_{10} particles varies as a function of one or several of the other variables, $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})$. We will use a linear regression model $Y_i = \mathbf{x}_i\beta + \varepsilon_i$ where the random errors ε_i are assumed to be pairwise independent and $N(0,\ \sigma^2)$. In order to fulfill these model assumptions we will have to use suitable transformations of both the emissions and some of the other variables.

Part 1. Particle emissions and fuel consumption

According to The Swedish Environmental Protection Agency (Naturvårdsverket), domestic transports is the largest source of PM_{10} particles, contribution 40 % of Sweden's emissions, with the second largest source being industry with a further 32 % of the emissions. The main particle sources are wear on tyres, road surfaces and brakes, construction and demolition industry, mostly road construction, and the use of fossil fuels in processes requiring high temperatures.

We might thus expect that the yearly fuel consumption could be used to explain some of the differences in particle emissions between the municipalities.

Lec.2 1(a). Motivate, with the help of suitable residual plots, why we should take the logarithm of the emissions:

$$\ln(\texttt{PM10}) = \beta_0 + \beta_1 x + \varepsilon.$$

Also determine whether we should use $x = \texttt{Fuel}$ or $x = \ln(\texttt{Fuel})$.

Lec.2 1(b). Use the model with $x = \ln(\texttt{Fuel})$ (*Model.1(b)*) and present a table with the β-estimates and their 95 % confidence intervals.

Plot $\ln(\texttt{PM10})$ against $\ln(\texttt{Fuel})$ together with this estimated linear relationship, its 95 % confidence interval and a 95 % prediction interval for future observations.

Also transform the relationship back to $\texttt{PM10} = \ldots$, and plot `PM10` against `Fuel` together with the estimated relationship, its 95 % confidence interval and 95 % prediction interval.

Comment on any problems with the model fit and how this is reflected in the behaviour of the residuals.

Lec.2 1(c). Express how, according to *Model.1(b)*, the expected emission of `PM10` particles would change in a municipality if the yearly fuel consumption would decrease by 10 %. Also calculate a 95 % confidence interval for this change rate.

Also calculate, with 95 % confidence interval, how much a municipality would have to reduce the fuel consumption in order to half its PM_{10} emissions.

Lec.4 1(d). Test if there is a significant linear relationship between the log-PM_{10} emissions and the log-fuel consumption, according to *Model 1(b)*. Report the type of test you use, the null hypothesis, the value of the test statistic and its distribution when the null hypothesis is true (including the degrees of freedom), the P-value of the test and the conclusion.

Lec.6 1(e). Determine how much of the variability in log-PM10 between municipalities this model can explain.

Part 2. Adding more explanatory variables

Lec.4 2(a). Now we turn to the other numerical variables. Plot log-PM_{10} against each of them and determine which of these should also be log-transformed. Any variable where a relative change is more natural than an additive change, e.g. any economic variable, as well as non-negative positively skewed variables are likely candidates.

R-code-tip: First pick out the observation identifier `Kommun` and the numerical variables from Fuel through PM10 with `select()`.

Then transform the resulting data frame with `pivot_longer()` (from the package `tidyr` which is included in the package `tidyverse`) so that all values for variables Fuel through GRP are placed in one long variable with default name `value` and the new variable `name` indicates which original variable each value came from. The `Kommun`- and `M10`-values are repeated a sufficient number of times and retain their old names.

You can then plot PM10 against each of the other numerical variables in separate subplots using `facet_wrap()` where the `"free_x"` option lets `ggplot()` pick different limits on the different x-axes:

```
df |> select(Kommun, Fuel:PM10) |>
  pivot_longer(cols = Fuel:GRP) |>
  ggplot(aes(x = value, y = PM10)) + geom_point() +
    facet_wrap(~ name, scales = "free_x") +
    scale_y_log10()
```

Lec.4 2(b). Plot the (log-transformed) numerical x-variables against each other and determine which of them have strong (at least ± 0.7) pair-wise correlations with each other.

R-code tip: Use `select() |> pivot_longer()` in order to place the x-variables in one long variable as before, and use `mutate()` to get the log-values instead.

Then transform the data back to wide format with `pivot_wider()`. The option `id_cols` is used to indicate which values come from the same municipality, `names_from` denotes where to get the variable names from and `names_prefix` adds a prefix to the original variable names, indicating that we now have log-values, while `values_from` indicates where the log-values come from.

Remove the factor variable `Kommun` (we do not want to plot it) and plot all pairs of log-values with `ggpairs()` from the package `GGally` (you may have to install it first). You also have to add, and run, `library(GGally)` at the start of your code.

```
#library(GGally)
df |> select(Kommun, Fuel:GRP) |>
  pivot_longer(cols = Fuel:GRP) |>
  mutate(value = log(value)) |>
  pivot_wider(id_cols = Kommun, names_from = name,
              names_prefix = "log", values_from = value) |>
  select(-Kommun) |>
  ggpairs()
```

Lec.4 2(c). Ignore any potential problems with multicollinearity and fit a model for log-PM_{10} as a linear function of all seven log-transformed numerical variables and present a table of the β-estimates and their standard errors.

Calculate the VIF-values for the x-variables and add them to the table. Determine whether it is reasonable to use all seven x-variables in the model or if some of them have issues.

Motivate which variable we should exclude first, refit the model without it (*Model 2(c)*) and add the new β-estimates for the remaining x-variables to the table, together with their new standard errors and VIF-values.

Compare the sizes of the standard errors for the β-estimate for the variable that had the second largest VIF-value, before we removed the one with the largest VIF-value. Explain the difference in size between the two standard errors and relate it to the relevant VIF-value.

Lec.4 2(d). Test if there is still a significant, at level $\alpha = 5\,\%$, linear relationship between the log-PM_{10} emissions and the log-fuel consumption, when we have added another five x-variables to the model, according to *Model 2(c)*. Report the type of test you use, the null hypothesis, the value of the test statistic and its distribution when the null hypothesis is true (including the degrees of freedom), the P-value of the test and the conclusion.

Also test if there is a significant, at level $\alpha = 5\,\%$, linear relationship between the log-PM_{10} emissions and at least one of the added five x-variables, according to *Model 2(c)* compared to *Model 1(b)*. Report the type of test you use, the null hypothesis, the value of the test statistic and its distribution when the null hypothesis is true (including the degrees of freedom), the P-value of the test and the conclusion.

Finally, determine how much of the variability in log-PM10 between municipalities *Model 2(c)* can explain.

Lec.4 2(e). Turn the categorical variable `Part` into a factor variable and present the number of observations in each of the three parts.

Add `Part` to *Model 2(c)*, producing *Model. 2(e)*, and present a table with the resulting β-estimates, their standard errors and suitable versions of the corresponding (G)VIF-values, with Df-values.

What is the reference category here? Is this a suitable reference considering the number of observations? Are there any multicollinearity issues?

Test if there are any significant differences in log-PM10 between any of the three parts of Sweden, when we have taken the six numerical variables into account. Report the type of test you use, the null hypothesis, the value of the test statistics and its distribution when the null hypothesis is true (including the degrees of freedom), the P-value of the test and the conclusion.

Lec.4 2(f). Add all two-way interactions between `Part` and each of the six numerical variables to *Model* **??**, producing *Model 2(f)*.

R-code tip: `(log(Fuel) + [etc] + log(GRP))*Part` gives all pairwise interactions involving `Part`.

Test if any of the added interactions is significantly, at level $\alpha = 5\,\%$, different from zero. Report the type of test you use, the null hypothesis, the value of the test statistics and its distribution when the null hypothesis is true (including the degrees of freedom), the P-value of the test and the conclusion.

This large model with everything contains a number of unnecessary variables. Our next step will be to eleminate some of them.

Part 3. Model validation and selection

Lec.5 3(a). Calculate the leverage (hat values) from *Model 2(f)* and plot them against the linear predictor, adding suitable horizontal lines as visual reference. Identify the three municipalities with the highest leverage and highlight them in the plot.

Add their names to the plot using the geometry `geom_text()`:

```
geom_text(data = filter(...), aes(x = ..., y = ..., label = Kommun))
```

Determine what makes each of these municipalities have such high leverage.

Hint: Plot pairs of x-variables against each other, separately for each factor category, `facet_wrap(~Part)`.

Lec.5 3(b). Calculate Cook's distance from *Model 2(f)* and plot them against the linear predictor, adding suitable horizontal lines as visual reference. Identify the three municipalities with the highest Cook's distance and highlight them in the plot. Are these the same municipalities that had high leverage?

Investigate the DFBETAS to find the β-parameter(s) that the municipalitiy with the highest Cook's distances had a very large ($> F_{0.5}$) influence on.

Then plot log-PM10 against the correspondingx- variable(s), highlighting the influential municipality, and explain why it had a large influence on that parameter.

Lec.5 3(c). Calculate the studentized residuals, r_i^*, for *Model 2(f)* and plot them against the linear predictor, using suitable horizontal lines as visual guides. Highlight the three municipalities with the highest Cook's distance. Do they also have large residuals?

Also identify the municipalities were $|r_i^*| > 3$ but which did not have high Cook's distance.

Plot $\sqrt{|r_i^*|}$ against the linear predictor, adding suitable reference lines and comment on the result, e.g., does the variance appear to be constant?

Also make a QQ-plot of r_i^* and comment on the result.

According to Naturvårdsverket, domestic transports is the largest source of PM_{10} particles, contributing 40 % of Sweden's emissions. The vast majority of this is due to wear on tyres, road surfaces and brakes. Burning wood for heating individual houses contributes 13 % of the emissions while burning biomass for production of electricity and district heating only contributes 3 %. Agriculture contributes 8 % of the emissions. The second largest source of Sweden's PM_{10} emissions is industry, contributing 32 % of Sweden's emissions. Of this, 69 % comes from the the construction and demolition industry, mostly due to road construction, while 16 % comes from paper and pulp processing (see Table 2 for a list of large Swedish paper mills and their locations).

A large part of the PM_{10} emissions from the rest of the industry is related to the use of fossil fuel in processes requiring high temperatures. These processes usually also result in high emissions of carbon dioxide, CO_2. In 2021, Swedish television, SVT, listed the 15 companies in Sweden that emitted the most carbon dioxide, during 2020 (see Table 1). Together they stood for 25 % of Sweden's CO_2 emissions. Note that the paper and pulp companies do not appear on this list, since they mainly use biomass as energy source.

In this project, we are more interested in the PM_{10} emissions per capita in general, and not so much if a small municipality happens to have a large PM_{10} emitting company within its borders.

Lec.5 3(d). Use Table 2 and 1 to find the common denominator between the three municipalities that have the largest Cook's D. Also keep in mind that they are fairly small municipalities (population wise).

We are trying to model the PM_{10}-emissions in "normal" municipalities so it would be best to remove the three from the data before we continue.

Create a new data set where the three municipalities with the highest Cook's D have been removed and refit *Model 2(f)* (numerical and interactions with Part).

Calculate Cook's distance and studentized residuals for the refitted Model 2(f) on the reduced data set. Also redo the plots of Cook's D against (the new!) linear predictor values (remember that the reference values also change), as well as the residual plots, identifying the municipalities where the new $|r_i^*| > 3$.

Identify the source of the high emissions in these municipalities. In addition to Table 2 and 1, also try Wikipedia. The municipality pages usually have a list of large companies. Some hints: The Swedish Air force wing F-7 Såtenäs is located in Lidköping and the Port of Gothenburg (= Göteborg) is the largest in Scandinavia.

Also discuss how well the assumptions of normality and constant variance of the residuals are fulfilled now.

Lec.6 3(e). Continue with the reduced data set and fit the null model using only an intercept as well as refitting *Model 1(b)* (log-Fuel) and *Model 2(c)* (numerical variables).

Perform a stepwise selection using AIC as criterion, and another using BIC, both starting with *Model 2(c)* and using the null model as lower scope and *Model 2(f)* as upper scope. Report the β-estimates and their standard errors for the resulting two models.

Lec.6 3(f). Do we need three Parts? Replace the variable Part, and the interactions, with a new version, NewPart, that separates one of the three parts from the other two. Base your choice on results of the stepwise models.

Fit a new version of *Model 2(f)* using Newpart instead of Part, *Model 3(f)*, perform the AIC and BIC stepwise selections with this model as upper scope instead and present the resulting two models. Discuss any interesting differences compared to 3(e).

Lec.6 3(g). Construct a table containing the number of parameters, the residual standard deviation s, the R^2, adjusted R^2, AIC and BIC for the following eight models: the null model, the refitted *Model 1(b)*, the refitted *Model 2(c)*, *Model 2(f)*, the AIC model and the BIC model from 3(e), and the AIC and BIC models from 3(f).

State which model you find best, and the reasons for your choice.

How much of the variability of the log-PM_{10} emissions can be explained by only the fuel consumption and how much is explained by the best model? Discuss whether the model make sense, e.g., if the variables included seem reasonable and if their β-parameters have the expected sign. Try to explain why you would expect the repationships to have the "expected" signs.

End of Project 1

	Company	CO_2	Description and locations
1	SSAB	4 777 065	Steel company. Production plants in Luleå (blast furnace, swe: *masugn*), Borlänge, Oxelösund (blast furnace) and Finspång.
2	Cementa AB / Heidelberg Materials	1 900 881	Building materials company with production in Gotland (Slite), and in Skövde.
3	Preem AB	1 538 189	Petroleum and bio-fuel company with oil refineries in Lysekil and Göteborg.
4	Luossavaara-Kiirunavaara AB (LKAB)	649 747	State owned mining company. Mines iron ore in Kiruna and Gällivare (Malmberget) with ore processing plants in Kiruna (Kiruna and Svappavaara) and Gällivare (Malmberget). Owns part of SSAB.
5	St1 Refinery AB	500 033	Oil refinery in Göteborg.
6	E.ON Värme Sverige AB	372 870	District heating company. District heating plants in Malmö and several other places.
7	Stockholm Exergi AB	361 050	District heating company with plants in Stockholm.
8	Borealis AB	345 364	Chemical company producing polyethylene in Stenungsund.
9	Boliden Mineral AB	279 876	Mining company. Mines zink and copper (and gold, silver and lead) in Hedemora (Garpenberg), Gällivare (Aitik) and Lycksele, Norsjö and Skellefteå, (Skelleftefältet).
10	SMA Mineral AB	278 960	Lime (swe: *kalk*) company. Several lime stone quarries. Production plants for quicklime in Rättvik (Boda, also producing slaked lime), Söderhamn (Sandarne), Luleå, and Oxelösund.
11	Tekniska verken i Linköping AB (publ)	266 807	District heating/electricity company with plants in Linköping and neighbouring areas.
12	Kubikenborg Aluminium AB (Kubal)	245 204	Aluminium company with a smelting plant in Sundsvall.
13	SYSAV	243 707	Garbage/waste/recycling company for south Skåne with a waste-to-energy plant in Malmö.
14	Renova AB	206 356	Garbage/waste/recycling company for west Sweden with a waste-to-energy plant in Göteborg (Sävenäs).
15	Nordkalk AB	188 926	Lime company. Several lime stone quarries. Production plants for quicklime in Gotland (Storugns) and for slaked lime in Luleå and Landskrona, and a plant for both quicklime and slaked lime in Köping.

Source: SVT, www.svt.se/nyheter/inrikes/15-foretag-...

Table 1: The 15 largest carbon dioxide emitters in Sweden 2020

Company	**Locations**
Billerud AB	Grums (Gruvön$^{(7)}$), Gävle$^{(7)}$, Lindesberg (Frövi$^{(5)}$)
Metsä Board Sverige AB	Örnsköldsvik (Husum$^{(7)}$)
SCA	Timrå (Östrand$^{(7)}$)
Smurfit Kappa	Piteå$^{(5)}$
Stora Enso	Hammarö (Skoghall$^{(7)}$), Älvkarleby (Skutskär$^{(5)}$)
Södra Cell	Mönsterås$^{(7)}$, Varberg (Värö$^{(7)}$), Karlshamn (Mörrum$^{(5)}$)

Capacity: (5) = 500 000–700 000; (7) = more than 700 000

Source: www-skogen.se/skogssverige/papper/massa-...

Table 2: Large paper mills in Sweden